Seasonal variation of soil respiration in restored upland peatlands

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Abstract

Peatland restoration by ditch blocking is widely used to raise water tables and reduce carbon losses, but its effect on soil respiration over the seasons is poorly known. We measured soil CO_2 efflux monthly for two years at three restored and three unrestored sites on a blanket bog in the northern uplands. Mean annual efflux was 31% lower at restored sites, and the difference was largest in late summer when water tables at unrestored sites fell below 25 cm. Temperature explained most of the seasonal pattern, while water-table depth controlled its amplitude.

Keywords: peatland restoration; soil respiration; water table; carbon flux

1. Introduction

Northern peatlands store about one third of global soil carbon while covering only a small fraction of the land surface. Drainage for grazing and forestry has lowered water tables across large areas, exposing peat to oxygen and accelerating decomposition.

Blocking drainage ditches is the most common restoration method because it is cheap and quickly raises the water table near the ditch. However, most studies report annual budgets and say little about how restoration changes the timing of carbon losses within the year.

In this study we ask whether restoration reduces soil respiration evenly through the year or mainly in the warm and dry months.

2. Methods

2.1 Study sites

The study was carried out on a blanket bog at 420 to 510 m above sea level with a mean annual rainfall of 1,840 mm. Six sites were selected:

- three sites where ditches were blocked with peat dams in 2017;
- three nearby sites where ditches remain open;
- all sites on peat deeper than 1.5 m with similar vegetation.

2.2 Flux measurements

Soil respiration was measured on vegetation-free collars with a portable infrared gas analyser.[a] Each site had five collars, and measurements were made between 10:00 and 14:00 on one day per month.

Water-table depth was recorded in dipwells next to each collar, and soil temperature was measured at 5 cm depth.

a Model EGM-5 with SRC-2 chamber; collars were inserted 5 cm into the peat six weeks before the first measurement.

Table 1. Site characteristics (means of three sites).

Variable	Restored	Unrestored
Water table (cm)	-8.4	-21.7
Peat depth (m)	2.3	2.1
Soil temperature (°C)	8.9	9.4
Bulk density (g cm^{-3})	0.11	0.13

2.3 Statistical analysis

We fitted linear mixed models with log-transformed efflux as the response, treatment and month as fixed effects, and site as a random effect. Temperature sensitivity was described with an exponential model and its Q10 value.

Table 2. Seasonal mean soil CO_2 efflux at restored and unrestored sites, 2024–2025.

Season	Restored efflux (g CO_2 m^{-2} d^{-1})	Unrestored efflux (g CO_2 m^{-2} d^{-1})	Difference (%)	p
Winter (Dec–Feb)	0.62	0.71	-13	0.21
Spring (Mar–May)	1.48	1.93	-23	0.04
Summer (Jun–Aug)	3.05	4.87	-37	<0.001
Autumn (Sep–Nov)	1.71	2.39	-28	0.01
Annual mean	1.72	2.49	-31	<0.001

3. Results

Soil respiration followed a clear seasonal cycle at all sites, with the lowest values in January and the highest in August. Restored sites had lower efflux in every season, but the difference was significant only from spring to autumn (Table 2).

The Q10 value did not differ between treatments, which suggests that restoration changed the amount of peat exposed to oxygen rather than the temperature response of decomposition. Efflux increased sharply once the water table fell below about 25 cm, a threshold reached only at unrestored sites.

4. Discussion

Our results show that the benefit of ditch blocking is concentrated in the warmest and driest part of the year. This matters because climate projections for the region predict longer summer droughts, when unrestored peat will lose carbon fastest.

Annual budgets based on a few summer campaigns may therefore overestimate the effect of restoration if they are scaled up without seasonal weighting. Monthly sampling, as used here, gives a more reliable estimate.

5. Conclusions

Ditch blocking reduced soil respiration by almost one third, mainly by preventing deep summer drawdown of the water table. Restoration programmes should give priority to sites where summer water tables fall furthest.

References

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